



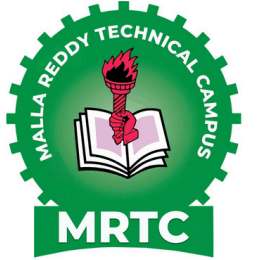
MALLA REDDY TECHNICAL CAMPUS

A Constituent Unit of MALLA REDDY VISHWAVIDYAPEETH

Deemed to be University

Approved by UGC & AICTE New Delhi

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STUDENT RECORD LOOKUP SYSTEM

Using Binary Search • Design and Analysis of Algorithms

A simple web-based system for fast student record lookup and algorithm visualization.

DAA Internal Project | Python + Flask + JSON

1. Problem Statement

Why do we need an efficient student lookup system?

- A university may maintain a large collection of student records identified by unique roll numbers.
- The system should determine whether a roll number exists and retrieve the corresponding student information.
- Searching every record one by one becomes less efficient as the number of records increases.
- The project demonstrates how Binary Search can reduce the number of comparisons when the records are sorted.

Dataset

50 student records
Roll numbers 101–150

2. Objectives

What the project is designed to demonstrate

Fast Lookup

Find a student by roll number using Binary Search.

DAA Concept

Demonstrate low/high/mid pointer movement and comparisons.

Algorithm Analysis

Compare Binary Search with Linear Search using complexity and counts.

Interactive UI

Search manually or use quick-search buttons for demo cases.

Trace Visualization

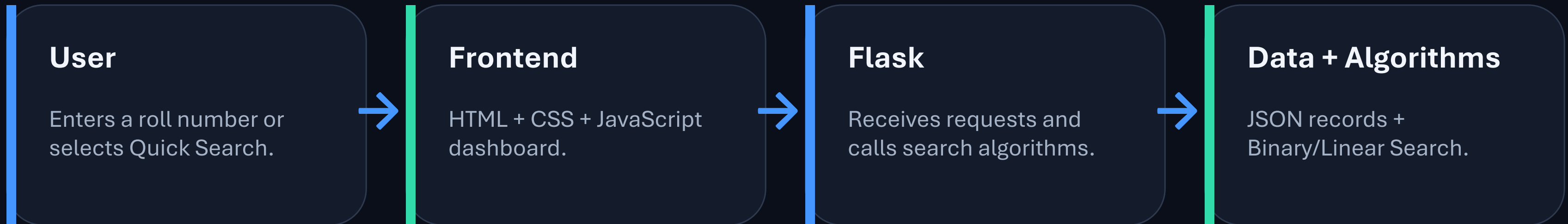
Show each Binary Search step with comparison and direction.

Input Validation

Handle empty, non-numeric, negative, and missing roll numbers.

3. System Architecture

Simple architecture with no database or unnecessary dependencies



Project Structure

app.py • algorithms/binary_search.py • algorithms/linear_search.py • data/students.json •
templates/index.html • static/style.css • static/script.js

4. Binary Search — How It Works

The main DAA algorithm used by the system

- The student records are kept sorted by roll number.
- Set two pointers: low at the beginning and high at the end.
- Calculate the middle position: $\text{mid} = (\text{low} + \text{high}) // 2$.
- Compare the target roll number with the middle roll number.
- If equal → record found. If target is smaller → search the left half. Otherwise → search the right half.
- Repeat until the record is found or the search range becomes empty.

Best Case

$O(1)$

Target is found at the first comparison.

Average / Worst

$O(\log n)$

The search space is repeatedly halved.

Space

$O(1)$ for the iterative implementation.

5. Features & User Interface

What the working application provides

Search Engine

Enter any roll number and retrieve the student record.

Quick Search

Demo buttons for 104, 125, 150 and 999.

Student Details

Displays roll number, name, branch, year and email.

Binary Search Trace

Shows low, mid, high, comparison and search direction.

DAA Analysis

Runs Linear Search and Binary Search for comparison.

Validation

Handles invalid and non-existent inputs gracefully.

6. Binary Search vs Linear Search

DAA complexity and practical comparison

Aspect	Linear Search	Binary Search
Method	Checks records one by one	Repeatedly halves sorted range
Best case	$O(1)$	$O(1)$
Average case	$O(n)$	$O(\log n)$
Worst case	$O(n)$	$O(\log n)$
Requirement	No sorting requirement	Data must be sorted
Implementation	Simple sequential loop	low / mid / high pointers

Project Comparison

7. Demo Results

Test cases used in the working application

104

✓ Student Found
Sneha Rao
IT • 3rd Year
Binary Search trace
shown.

125

✓ Student Found
Valid record
Demonstrates a
successful lookup.

150

✓ Student Found
Valid record
Search reaches the record
through the sorted range.

999

✗ Not Found
No matching record
Binary Search eliminates
ranges until empty.

What the demo proves

The system is not only displaying a webpage: it performs the search, returns the matching JSON record, records Binary Search steps, and supports algorithm comparison.

8. Conclusion & Future Scope

Key takeaway from the project

Conclusion

- Successfully implements student lookup using Binary Search.
- Visualizes the algorithm step by step.
- Demonstrates the difference between $O(\log n)$ and $O(n)$.
- Provides a simple interactive interface suitable for demonstration and viva.

Future Scope

- Connect to a real database for larger datasets.
- Add sorting and record-management features.
- Add authentication for authorized users.
- Extend the system with more advanced analytics or reporting.

Final Message